

# CONSULTATION



The following is the Consultation note from Amanah Medical. Requested by our Partner Facility for evaluation of persistent altered mental status as well as persistent Hypernatremia.

**Dr. Ahmed Mustafa Abbasi, Internal Medicine Specialist.**

## History of Presenting Illness

The patient is a XX old xxxx with a past medical history of cognitive impairment, COPD (confirmed w PFTs), Heart Failure with Preserved Ejection Fraction (EF 50–55% 06/2026) and BPH who presented to Partner Hospital in Acute Respiratory Distress from his home. Patient was evaluated in the ER and found to be hypotensive, hypoxemic and altered. He was intubated in the ER to protect his airway and was subsequently transferred to the ICU for further evaluation. While the patient was in the ICU he was hypotensive requiring Norepinephrine as well as Vasopressin. His respiratory status declined despite mechanical ventilation and he was found to have ARDS 2/2 to Streptococcus Pneumonia. Hypoxemia did not respond despite intensive care team's vent management and patient was persistently hypoxic in the 70–80 range until he was eventually prone. Saturation slowly improved and patient was extubated to room air. 3 days post presentation patient developed Hypernatremia. Presumed fluid losses and NGT was placed with frequent free water flushes. 4 days post extubation patient has not returned to his mental baseline and has a persistent hypernatremia between 150–154. Amanah Medical Services were requested for evaluation on day 5 post extubation.

### 1. Subjective

Reviewed video documentation from the Primary Team, patient is spontaneously responsive but dysarthric and seems to be in distress.

### 2. Objective

**General:** Quite Frail Appearing, Moaning from pain and struggling to articulate

**Cardiovascular:** Telemetry monitoring consistent with sinus rhythm and PVCs

**Respiratory:** Saturating 95% on RA despite NGT occluding left nostril, Non Labored Respirations

**Gastrointestinal:** Distended and quite tender to palpation when evaluated by primary team

**Genitourinary: Foley catheter in place, draining light yellow urine**

**Neurological: Dysarthric, moving extremities spontaneously**

**Extremities: No Significant Edema on exam by Primary Team.**

- Intake and Output: Review of the chart indicates that patient has been on Free water flushes, Tube feeds as well as D5W at 150cc/hr for 72hrs. During this time period he has received in excess of 10 Liters of Free water, not accounting for IV medications. His urine output has been surprisingly high at 4–6L/day in the absence of diuretics
- Laboratory Findings: Patient had a baseline Sodium of 138 when he came to the hospital initially and stayed relatively stable until Hospital day 3, whereupon he had a significant increase of his sodium to 150 and the next day 154. During the subsequent 4 days he received 10L of Free water with minimal improvement of his sodium to 150. Review of his Urine Analysis on Admission demonstrated a urine specific gravity of 1.017, no LE, Nitrites etc, of note, at the time the patient had a sodium of 138.

### **3. Assessment and Plan**

**#Acute Metabolic Encephalopathy (2/2 Persistent Hypernatremia)**

**#Possible Ischemic injury**

- I suspect that patients residual metabolic encephalopathy is multi factorial in origin. It is difficult to truly establish the level of decline from his baseline until his sodium has been adequately corrected. Refer to below. If sodium is corrected with the below pt will likely benefit from MRI brain w/o contrast to evaluate for Cerebral Ischemia in addition to the below to rule out hypoxic injury contributing to his presentation.

–Correct Sodium and check MRI brain w/o contrast pending below evaluation.

**#Persistent Hypernatremia**

**#Suspect Partial Central Diabetes Insipidus Associated with above**

- This is a very interesting presentation of Diabetes Insipidus. Patient had no history of DI when presenting to the hospital and demonstrated both a normal sodium and an appropriate Urine Specific gravity for his level of sodium. He was both hypotensive and hypoxemic for an extended period of time during initial stabilization despite pressors and mechanical ventilation. At that time he demonstrated adequate ability to maintain sodium homeostasis. Starting Hospital day 4 patient had a sudden rise in his sodium that has not come back down. His calculated free water deficit is 1.2L. Patient received 10X his free water deficit with no improvement of his sodium levels. During this time period he has had 4–6L UO per night. This suggests that the patients Collecting ducts are malfunctioning. Under a normal physiologic state the body should not waste such volumes of free water while hypernatremic. This is suggestive of DI. I do not see any reason for why the patient may have a nephrogenic DI so suddenly. This could be a partial Central DI 2/2 to hypoxic injury to the hypothalamus. The reason I believe the presentation was delayed is that the hypothalamic injury is depleting the production of ADH but there is still residual ADH production from

prior being released from the posterior pituitary. The presentation may also have been delayed by Vasopressin?

- Check Urine Analysis, If specific gravity is not elevated suggestive of poor concentrating ability
- Check Urine Osm, if this is not elevated in the 800–900 range this is atypical in the setting of hypernatremia and hyperosmolality. Suggestive of impaired free water reabsorption at least partial if not complete central DI.
- Check Serum Osm to compare directly to urine osm
- Post above testing trial DDAVP and reassess above. If consistent with Partial Central DI, evaluate hypothalamus/posterior pituitary via MRI Brain, will also show if underlying hypoxic brain injury.

## **SIGNATURE**

**Dr. Ahmed Abbasi**

**Internal Medicine–ABIM**